Cyber Threat Defender

Instructor guide · Req 8b competition · 30 minutes (2:14-2:44)

What Req 8b asks

Participate in a cybersecurity competition with members of your troop, school or another counselor-approved group, either one you design or one on an existing platform. Cyber Threat Defender (CTD) is an existing platform from the UTSA Center for Infrastructure Assurance and Security, free for K-12 education, for ages 11 and up.

Test before class (required)

I could confirm that CTD is a free download, that it runs on Windows PC and Mac through the CIAS Launcher, that the digital game plays against computer opponents, and that CTD's own rules apply (below). I could **not** confirm how long a digital round takes, whether the download needs an account, or whether 25 devices can run it at once. Install it on a Mac and a Windows device, play a full round, and time it. Page: cias.utsa.edu/ctd/play-online. Launcher and games page: cias.utsa.edu/k-12/cybersecurity-games. Questions: cias@utsa.edu.

Launch	Play	Wrap
4 min	21 min	5 min

PREREQUISITE: install before class

Every device that will play must have the CIAS Launcher and Cyber Threat Defender installed and tested before class starts. There is no time to install during the 30 minutes. Install on your own devices and any spares, and ask families to install at home (email below).

Before class

1) Get counselor approval for CTD as the Req 8b competition. 2) Install the CIAS Launcher and CTD on your own device and play. 3) Send the family email below at least a week ahead so Wardens install it at home (Mac and Windows). 4) Print page 2 (one lodge slip per lodge) and the scoring sheet. 5) Plan B: lodges share one laptop and Wardens take turns, or play on your devices.

Run it

Launch (4 min): Wardens open CTD and start a game against the computer. **Play (21 min):** walk the room. Ask each Warden to name one defense card they used and the real-world defense it matches (backups, a patch, MFA, a firewall). **Wrap (5 min):** Wardens report their points; lodges add them; ask which defenses mattered most and which attacks surprised them.

Scoring (Cup of the Silver Padlock)

Lodge total = the sum of its Wardens' points at time-up (a game does not have to finish). Cup points: 1st **+5**, 2nd **+3**, 3rd **+1**. Each Warden who explains a defense card in real terms: **+1**. Counselor breaks ties.

Family email (copy and paste)

Subject: Cybersecurity merit badge: install a free game before class

Hello! At our class on [date] we will play Cyber Threat Defender, a free cybersecurity card game from the University of Texas at San Antonio. Please help your Warden install it before class on a laptop (Windows or Mac): go to cias.utsa.edu/ctd/play-online and follow the steps for the CIAS Launcher. Please play one round so you know how it works. Nothing else is needed, and no personal information should be entered. If it will not install, no problem: Wardens will share a device in class. Thank you!

Cyber Threat Defender: Quick Start

One per lodge · summary of CTD's published rules (cias.utsa.edu/ctd/rules)

Goal. Reach **30 points** while having more points than your opponent. Asset cards earn points. The more assets in your network, the more points you earn each turn.

Setup. Place one Asset – System (Desktop Computer) and one Asset – ISP (ISP Connection) face up. Draw seven cards.

A turn. Play up to **three** cards. Event cards can be played at any time and do not count toward the limit. At the end of your turn, discard down to your hand limit and draw two.

Card types. Asset cards build your network and earn points. **Attack** cards remove points or cards from an opponent. **Defense** cards counter attacks. **Event** cards can change a round. Read each card's text for what it can counter.

You lose if you must draw and your draw pile is empty, or if you reach −45 points.

Think like a defender

For each Defense card you play, say what it protects in real life: a backup, a software update, MFA, a firewall, a strong password. Your counselor will ask.

Safety

Use only the game. Do not enter personal information or accounts. Keep the volume low.

Lodge slip — cut and hand to the counselor at 2:44

Lodge: _____ Warden 1: ____ pts Warden 2: ____ pts Warden 3: ____ pts Warden 4: ____ pts Warden 5: ____ pts **Total:** _____

A defense card we used and the real-world defense it matches: _____

Cyber Threat Defender: Scoring Sheet

Counselor copy · add to the Cup of the Silver Padlock

Lodge	Warden pts (sum)	Rank	Cup pts (+5/+3/+1)	Defense explained (+1 each)	Notes	Cup total	
1							
2							
3							
4							
5							
6							

Req 8b check

☐ Counselor approved CTD before class ☐ Every Warden took part (own device or shared) ☐ Scores recorded and added to the Cup ☐ Sign-off tracker updated (8b)

Wrap-up questions

Which defense helped the most? Which attack caught you off guard? What real-world habit does that match? How is a game like this different from a real attack?

If devices fail

Pair Wardens. Use one laptop per lodge and rotate every 5 minutes. Score the lodge's best game. The competition still counts as long as every Warden participates.